



CO544

Machine Learning and Data Mining

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Feature Selection



- Selecting a subset of elements from a given set of candidate features
- An optimization technique that, given a set of M features as $X = \{x_i \mid i=1,2,\dots,M\}$ an attempt to select a subset of size N as $Y = \{x_{i_1}, x_{i_2}, \dots, x_{i_N}\}$ where $N < M$ that leads to the maximization of some criterion function without additional feature transformation or construction

Finding the optimal subset of features?

Sufficient features to achieve high recognition rates?

Feature Selection cont.



Feature Selection vs. Feature Extraction

Aspect	Feature Selection	Feature Extraction
Definition	Selecting a subset of relevant features from the original set	Transforming the original features into a new set of features
Purpose	Reduce dimensionality	Transform data into a more manageable or informative representation
Process	Filtering, wrapper methods, embedded methods	Signal processing, statistical techniques, transformation algorithms
Input	Original feature set	Original feature set
Output	Subset of selected features	New set of transformed features
Information Loss	May discard less relevant features	May lose interpretability of original features

Why Feature Selection?



- Features can be irrelevant or redundant
- Decrease problem dimensionality

For a fixed number of samples, the error of a classifier will increase (generally) when increasing the number of features

When the number of features is very large relative to the number of observations, certain algorithms might struggle to train effective models

Feature Selection is a two-way process

1. Define objective function (attach value to every subset of features)
2. Formulate algorithms

Optimal Feature Selection?



Goal?

- Processing an algorithm using an optimal feature subset to best process novel instances of domain data (unseen data)

Depends on?

- The objective
- Dataset
- Domain Knowledge
- Processing Algorithms
- Feature Selection Criterion
- Original set of features

Optimal Feature Selection cont.



Two paradigms

- Minimal Representation

Choose a minimal feature subset that fully describe all concepts

- Maximal class separability

Choose the feature subset that guarantees maximal between-class separability for a reduced dataset

Optimal Feature Selection cont.



- Selecting the optimal set of features is connected with **feature relevancy**
- Relevance of a feature is its ability to contribute to improving the classifier's performance
- Relevancy could be deterministic or probabilistic
- Feature selection methods give a dataset with reduced number of features according to a specific optimal criterion

Two methods, based on feature selection criterion used

- Open loop method
- Closed loop method

Open loop method



- Selecting features for which the reduced dataset maximizes between-class separability
- Do not consider the effect of selected feature subset on the performance of the processing algorithm
- No feedback mechanism from the predictor performance
- Features are selected on the basis of their scores in various statistical tests (correlation of the feature with the outcome)

Information contained on dataset alone!



Open loop method

- 
- Advantages?
 - Disadvantages?
 - Examples?

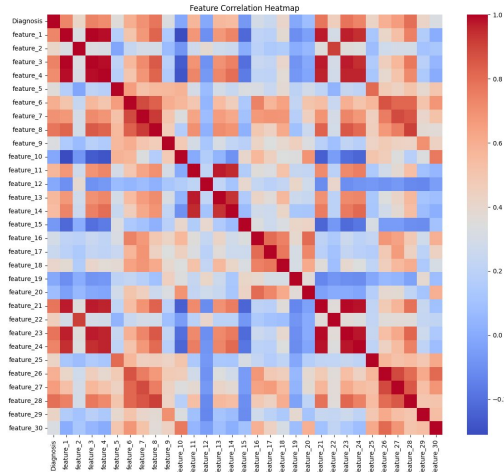
Open loop method



- Advantages
 - Fast execution
 - Higher generality
- Disadvantages
 - Usually select larger feature subset
- Examples
 - Filter Approaches

Open loop method cont.

Feature\Response	Continuous	Categorical
Continuous	Pearson's Correlation	LDA
Categorical	Anova	Chi-Square



Some other measures:

- Minimal Description length
- Mutual Information
- Inconsistency Count
- **Interclass Separability**

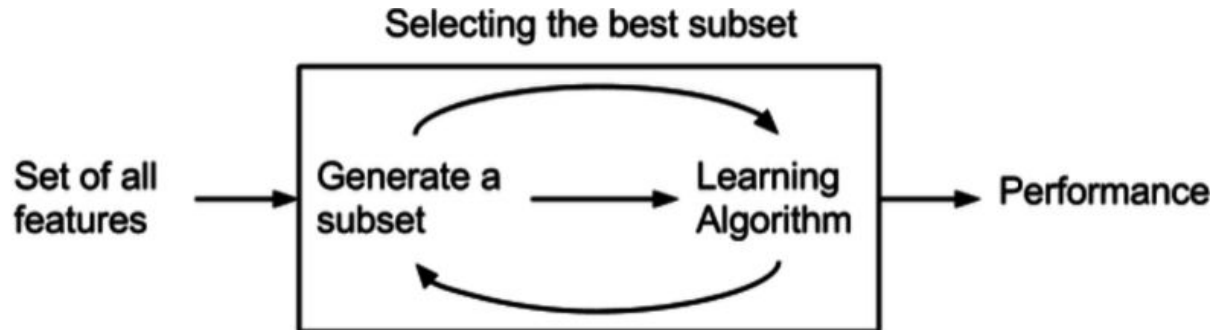
Closed loop method



- Selecting features based on the processing algorithm's performance as a criterion for feature subset selection
- The optimal subset of features is identified by evaluating the algorithm's performance
- Ultimate goal is the best performance of the processing algorithm
- Generally provide a better selection of feature subsets

Closed loop method cont.

- Criterion:
 - Count the number of misclassified patterns for a specified feature subset
 - Mean squared prediction error
 - Mean absolute prediction error



Closed loop method cont.

- 
- Advantages?
 - Disadvantages?
 - Examples?

Closed loop method cont.



- Advantages
 - Higher accuracy
 - Less overfitting
- Disadvantages
 - Slow execution
 - Computationally expensive
 - Low generality
- Examples
 - Wrapper Approaches
 - Forward Selection
 - Backward Elimination
 - Recursive Feature Elimination

Closed loop method cont.

- **Forward Selection:** Forward selection is an iterative method in which we start with having no feature in the model. In each iteration, we keep adding the feature which best improves our model till an addition of a new variable does not improve the performance of the model.
- **Backward Elimination:** In backward elimination, we start with all the features and removes the least significant feature at each iteration which improves the performance of the model. We repeat this until no improvement is observed on removal of features.
- **Recursive Feature elimination:** It is a greedy optimization algorithm which aims to find the best performing feature subset. It repeatedly creates models and keeps aside the best or the worst performing feature at each iteration. It constructs the next model with the left features until all the features are exhausted. It then ranks the features based on the order of their elimination.

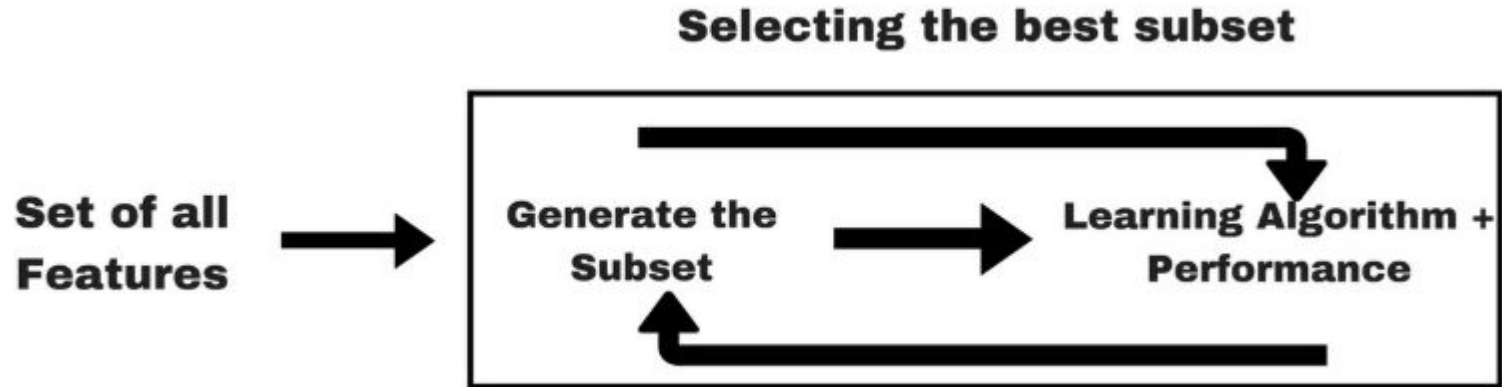
Closed loop method cont.



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Embedded Methods

- Performs feature selection as a part of model construction
- Combine the qualities of both filter and wrapper approaches



Feature Selection Process



1. Generation: Select feature subset candidate
2. Evaluation: Compute relevancy value of the subset
3. Stopping criterion: Determine whether subset is relevant
4. Validation: Verify subset validity

Feature Selection Process: Generation



- Can be started with no features or all features, or a random subset of features
- Subsequently add, remove features (or do both)
- How do we examine the feature space?
 - Complete/ exhaustive selection
 - Heuristic selection
 - Random selection
 - Rank selection

Feature Selection Process: Evaluation



- Measures the goodness (relevancy) of the selected subset of features
- Evaluation Functions
 - Distance
 - Information
 - Dependency
 - Consistency
 - Classifier error rate

Feature Selection Process: Stopping Criterion and validation



- Stopping Criterion: Evaluate whether the optimal subset based on some evaluation function is achieved or not
- Validation: Compare results with already established results or results from competing feature selection methods

Discussion Points



- Optimal vs Suboptimal feature selection methods?
- Feature Selection methods examples
 - Branch and Bound
 - Beam Search
 - Individual Feature Ranking

Branch and Bound Feature Selection



- Based on the assumption that a feature selection criterion satisfies the monotonicity relation
- It assumes that the addition of features can only increase the value of the objective function
- The underlying concept is **divide and conquer**
- Divide: Partitioning the entire set of features into smaller subsets
- Conquer:
 - Giving a bound for the best solution in the subset
 - Nodes whose objective functions are lower than the current best are not explored

Branch and Bound Example

$n = 4$
 $x = 2$

